

Crime Intention Detection System Using Deep Learning

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Abstract— Circuit Television Cameras (CCTV's) are widely used to control occurrence of crimes in the surroundings. Although CCTV's are deployed at various public and private areas to monitor the surroundings there is no improvement in the control of crimes. This is because CCTV requires human supervision which may lead to human prone errors like missing of some important crime events by human while monitoring so many screens recorded by CCTV's at same time. To overcome this issue, we came up with Crime Intension Detection System that detects crime in real time videos, images and alerts the human supervisor to take the necessary actions. To alert the supervisors or nearby police station about the occurrence of crime. We added SMS sending module to our system which sends SMS to concern person whenever crimes are detected. The proposed system is implemented using Pre-trained deep learning model VGGNet-19 which detects gun and knife in the hand of person pointing to some other person. We also compared the working of two different pre-trained models like GoogleNet InceptionV3 in training. The results obtained with VGG19 are more accurate in terms of training accuracy. This motivated us to use VGG19 with little fine tuning to detect crime intention in videos and images to overcome the issues with existing approaches with more accuracy. And we made use of Fast RCNN and RCNN these algorithms are well known as Faster RCNN this helps us to draw the bounding box over objects in images like person, gun, knife and some untrained images are marked with N/A. Algorithms help for detection and classifications of objects over images.

Keywords—CNN, Faster RCNN, RCNN, VGGNet19, GoogleNet Inception V3.

I. Introduction

Deep Learning pre-trained models are designed models which help for people to learn about algorithms or try out existing framework for better results without designing explicitly. And deep learning neural network has five layers including input and output layers with Convolution, Max-Pooling and Fully connected layer, functions of these layers are discussed later in further sections. Due to some restricted time, memory and resources like CPU, Processers so many people use to prefer Deep Learning pre-trained concepts. And these pre-trained models will give the best and accurate results compare to machine learning which makes us to design explicitly. There is lot of human intervention requires to detect weapons in surveillance videos which tends to human prone errors. By observing videos of large volume or maintaining two or more footages by persons, human guards may get tired or fall asleep and they missed out some rarely occurring crime intention scenes who maintain more than one footage. For this purpose, developing the automatic surveillance system is necessary that detect weapons with less computational

time to reduce crime occurrence. Crime Intention Detection System is designed to solve problems occurring at ATM, Bank and Public Places by detecting the weapons in hand of persons which are used detect the crimes before it occurs, the designed system uses the Deep Learning pre-trained models like GoogleNet and VGGNet-19. Following section of the paper will be discussing about the pre-trained models. And we proved VGGNet-19 model performs best in training accuracy, better classification and decision making compared to GoogleNet Inception V3 model.

II. LITRATURE SURVEY

A. Automatic Handgun Detection Alarms in Videos Using Deep Learning[1]:

In this paper author described an automatic handgun detection system in surveillance videos, System controls the occurrences of crimes by detecting the guns in the surveillance videos, it will classify objects like gun or not, predict the crime occurred or not, it gives the solutions which have been compared with sliding window proposal approach, the good results are obtained by FRCNN and RCNN based models trained on dataset. This model shows a high crime prediction occurrence even in low quality videos then produces the best satisfactory results.

B. Automatic Visual Rrecognition of Armed Robbery[2]:

In this paper author explained classic method to detect position of armed robbery, It detects the persons who holds knife in hands with different positions pointing to persons, it uses the skeleton silhouette algorithms which divides body part into different pieces by identifying position of reside arm in air holding a knife with different position in different angels are recognized and knife is detected.

C. Robust Object Detector Application to detection of Visual Knives[3]:

In this paper author explained visual method to detect an automated weapon like knife in hands, it makes use of novel object detection algorithms to detect the visual knife for the given video dataset. This approach is used to detect the knife rotations with its different scale and positions, which is a challenging task to detect the scaling and position of knife, the main problem is to detect the knife holding at different scale and position in given video dataset. And all possible knife detection with different scale and position is designed and implemented. It will extract features by using the foreground segmentation and makes use of FAST (Features from Accelerated Segment Test) for

feature detection and it uses MRA (Multi-Resolution Analysis) used for classification.

III. PROPOSED METHOD

Designed system is experimented with datasets for both videos and images that are collected from YouTube and Google, collected dataset is of type robbery, murder and some illegal activities holding weapons in hand, where as the use of weapons are strictly prohibited in areas like ATM, Bank and some public places.

The main aim of project is to detect the weapons in less training time with accurate results and reducing false conditions compared to machine learning techniques and to make CNN work without performance degradation with less number of training samples. Pre-trained models like GoogleNet and VGGNet-19 models are trained with million plus images which will detect the objects in new images with minimum errors. Based on high training accuracy we decided to use VGGNet19 model which will classify and detects objects accurately.

The detailed design of proposed system is shown in figure 3.1. Input frames are accepted through input layer where pre-processing occurs at input layer, preprocessed images are passed from Convolution, Max-pooling and FC layers where feature extraction, filtering of features, mapping of features and classification occur, output layer proceeds output if any crime intentions are detected it sends crime intentions security message using registered API otherwise not, evaluation of model is done with evaluation parameters based on training we got best rank accuracy for VGG19 compared to the GoogleNet Inception V3.

For SMS sending module we made use of registered API.

A. Prepare Dataset

Collect all possibilities of positive images of gun, knife with different sizes holding at different angles with different colours and load all images to folders of weapons (guns and knives) and collect negative images that look similar to the weapons like gun and knife to not weapons folder.

B. Pre-processing data

Input frames are feeded to Input layer ,this layer pre-process the given images of any size (like 256*256 width, height) converts to 224*224*3 standard size (RGB values)processed by subtracting RGB values from pixels.As shown in below figure 3.2.

C. Feature Extraction

Keras will help to extract feature from pre-trained weights of ImageNet model trained with possible large datasets, a layer extracts the feature pattern from pre-trained images of ImageNet. VGG19 which has 19 weights of layers to extract feature from ImageNet and k-means used to cluster weapons and non-weapons based on features extracted from images, extracted features are shown in figure 3.3.

As shown in below figure 3.3, -1 indicates feature extracted from the image where no features of weapons present and 1 indicates extracted feature of the weapon this is an example taken just for a reference and understanding purpose actual feature extraction is dependent on the machine.

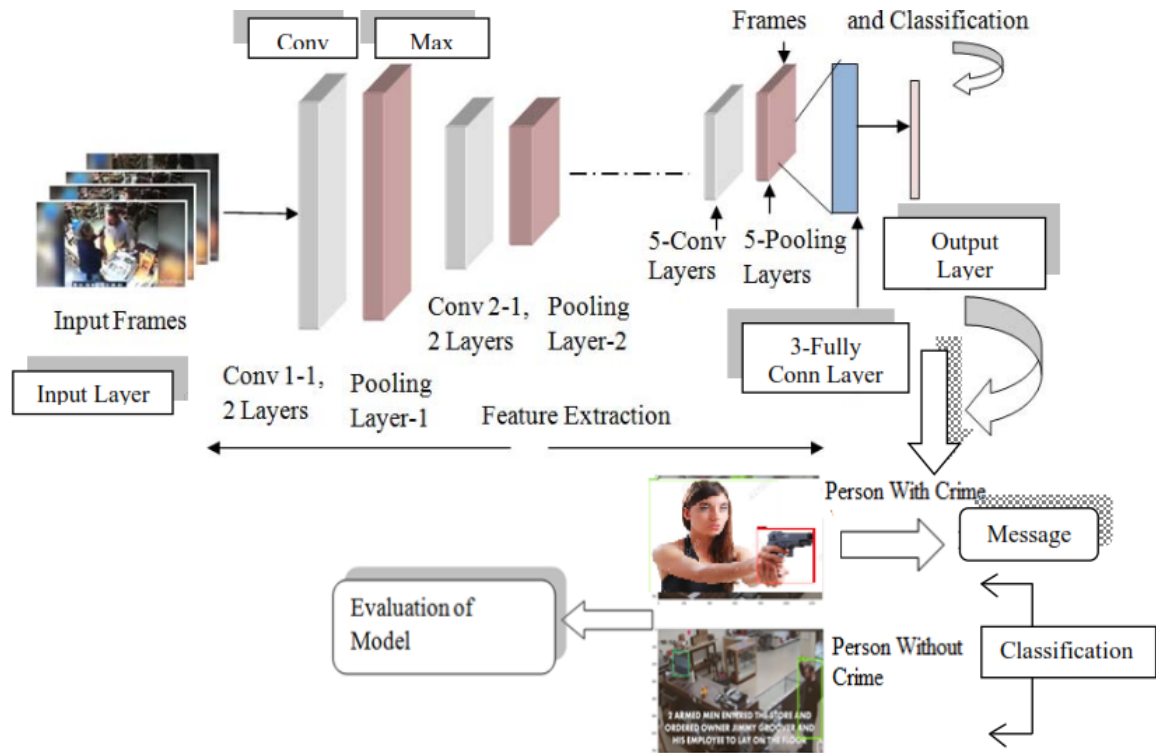


Figure.3.1 design of proposed system

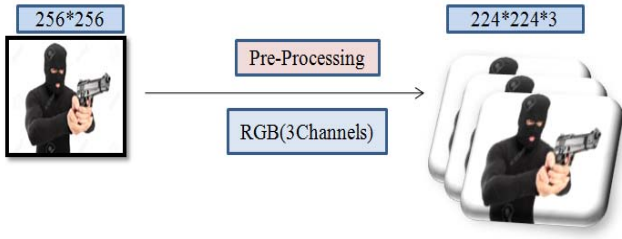


Figure.3.2 Pre-Processing

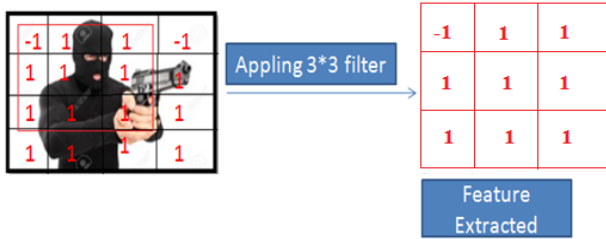


Figure.3.3 Feature Extractions

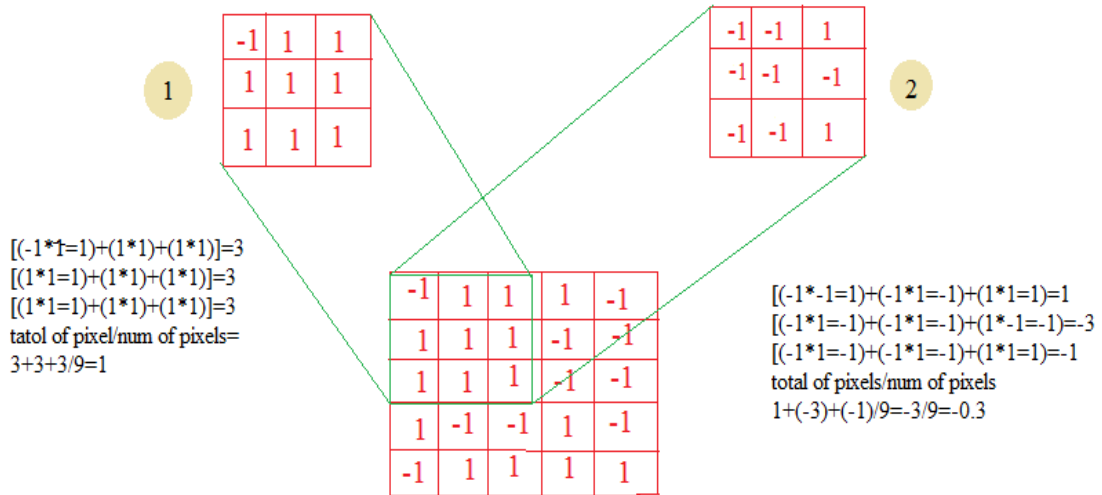


Figure.3.4 Feature Mapping

- Convolution layer not known's where are features to be extracted so this layer plots filter of 3*3 for all image pixels then calculates dot product of pixels values as $-1*-1=1$ and $1*1=1$ this means extracted feature pixels of testing images are very close to trained pixel values. If it as $-1*1=-1$ or $1*-1=-1$ values where matching of pixel will be very equal to negative values or 0 these are far from matching already trained pixels.
- Filtering process is carried out with all patches of images after completing all pixels later arrange features to two dimensional array contains all the possible features of images, calculated pixel values are given input to next layer(like 1,0.3, calculated above as dot product).

D. Feature Mapping

It is process used to match extracted features to trained features as shown in figure below 3.4. Extracted every filter is of form 3*3, after calculating the dot product values are like 0, 0.3, 0.4, ..., 0.85, 0.9, 1. If the values are near or equal to 1 then feature is matched to the trained ImageNet dataset which indicates the similar to the feature of weapons present. If the calculated dot product values are not nearer to 1 like 0, 0.3, 0.5 these indicates the weapon features are not present. Complete feature mapping shown in figure 3.4.

E. Max-Pooling Layer

Pooling layer applies Max-Pooling function to all images to shrink (reduce large number of pixels) and get very important features from the image. This process repeated to get small narrow depth extracted features from image shown figure 3.5 is only the one sub part of 3*3 filter extracted from image. Assume you extracted all the possible matching filters from the given testing images which are trained in the ImageNet datasets as shown in Figure 3.5 extracted features of objects are features of gun and knife. Below shown is only one sub part of weapon feature that we assumed for reference purpose how the pooling occur.

F. ReLU Layer

Small layer but makes important task changing negative values to 0 value in images, if layer reads any negative pixel values then make it to zero and remaining matched pixel positive values are assigned any positive values. There are two ReLU layers and another ReLU layer after convolution layer help for feature mappings In below shown diagram the activation function compares the trained model pixel value to new image pixel values, if function detects any negative value in pixels then set it to 0 otherwise left it as positive, In above example we consider 1 for matched feature and non feature values set -1 (1 and -1 are assumed values). A negative value indicates no similar feature of weapons present and positive value shows that some similar features of weapons present in the image.

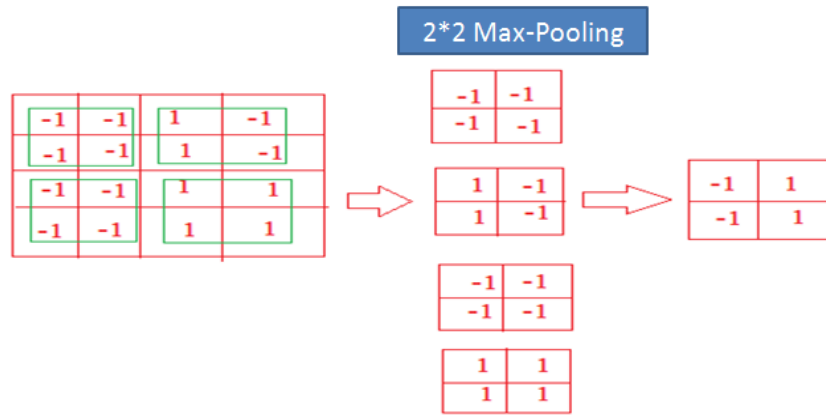


Figure.3.5 Maxpooling

It helps to calculate mathematical function of pixels and reduces the complexity of calculations. As shown below the figure 3.6 is just an example of activation function.

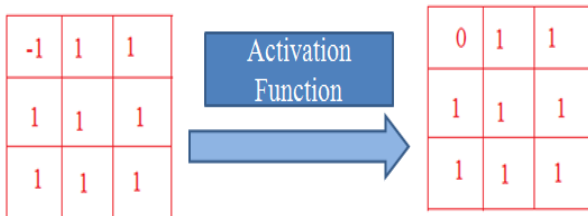


Figure.3.6 Activation by ReLU

G. Fully Connected Layer

It is final and the last layer contained 3 FC layers, first layer will identify whether trained object present in testing image or not, second layer will detect present objects in image and third layer will classify the objects for which class it belongs in our cases there are two classes one is weapon and other is no-weapon, if it detects the values nearer or equal to 1 then it belongs to class of weapons otherwise to 0 that is no-weapons classes. Which shown below in the Figure 3.7 which has single dimensional values.

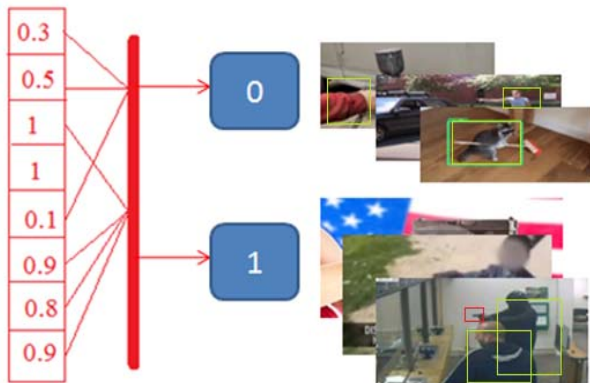


Figure.3.7 Classification by FC Layer

H. Localisation of Objects:

Bounding box is used to locate the objects like weapons, not weapons in images. To draws bounding box over images it used the Faster-RCNN which is obtained from combination of Fast RCNN and RCNN algorithms. Faster-RCNN will give us the high potential detection results even though for low quality YouTube videos and image dataset. It is totally based on the Regional-Proposal Network (RPN) used to detect the objects using two approaches.

IV. RESULTS AND DISCUSSIONS

A. Comparing VGGNET19 Model and GoogleNet Model

VGGNet19 give best training accuracy results with 100% for crime datasets compare to GoogleNet inception model gives only 87% accuracy and VGGNet19 classify the presence of objects in images at FC layer, this layer is good at classifications but GoogleNet classify objects at Max layer which is very mushy layer which perform pooling and classification at same layer, more accurate results will not be formed because it uses only one FRCNN algorithm but VGGNet-19 uses both FRCNN AND RCNN algorithms for localization of objects in images. And preprocessing is very good in VGGNet19 by converting raw dataset into standard data size with 224*224 sizes. Training accuracy of matrices obtained for trained dataset for VGGNET-19 is 100%, F1 score is good for VGGNet-19 always lies between the precision and recall as shown in below Table 4.1(a) for VGGNet19 and GoogleNet evaluation results are shown in Table 4.1(b).

Table 4.1(a)

Accuracy 100%	Training Accuracy of VGGNET19			
	Accuracy	Recall	F1-score	Support
0	0.93	0.90	0.95	30
1	0.91	0.93	0.92	110
Avg/Total	0.92	0.91	0.93	140

Table 4.1(b)

Accuracy 87%	Training Accuracy of GoogleNet V3			
	Accuracy	Recall	F1-score	Support
0	0.71	0.67	0.69	20
1	0.68	0.84	0.91	110
Avg/Total	0.69	0.75	0.8	130

B. Results for normal dataset:

Experiments are done on real time captured images, videos and results obtained from proposed method are demonstrated in figures below. In figures 4.2 (a), (b), (c) detection of objects other than weapons are shown in figures that is of no crime intentions are detected for which no SMS will be sent and our system performs well in detecting these normal objects which look similar to the objects like gun, knives.



Figure.4.2(a) No crime



Figure.4.2 (b) No Crime



Figure.4.2(c) No Crime

C. Results of Crime Datasets:

Figure 4.3 (a) (b) (c) (d) depicts the detection of crime by the proposed method. Both gun and knife are detected with red color which shows crime intentions are detected but person with green color, if any person detected with knife or gun in hands the SMS module activate and send SMS to concerned security person about crime intentions detection which can reduce crime before it occurs. And figure 4.3 (e) shows the crime intention detection system security message.



Figure.4.3(a) Crime Detection



Figure.4.3(b) Crime Detection



Figure.4.3(c) Crime Detection



Figure.4.3(d) Crime Detection

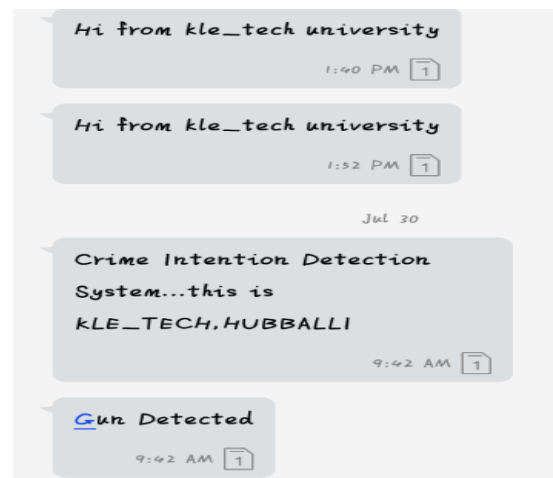


Figure.4.3(e) Crime Notification Message

V. CONCLUSION

Proposed Crime Intention Detection System is automated system which controls occurrence of crimes by detecting gun and knife in hands of a person using VGGNET19 pre-trained model in less computation time compare to GoogleNet Inception V3 model. If a person is detected with weapons like gun and knife then our system sends the crime intention detection security message to the register number. Proposed system gives good results compared to other existing approaches for crime detection.

The Designed Crime Intention Detection System features can be embedded to CCTV to detect the crime scenes like detection of gun, knife in hands of persons, if crime is to occur the added feature make the CCTV to send automatically the crime intention security messages to register number.

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